

# A SAMPLE PAPER

TEST AUTHOR

ABSTRACT. This is a short abstract. See Section ?? for details.

## 1. MAIN RESULTS

We begin with a foundational theorem, rooted in classical work [?, ?].

**Theorem 1.1.** *For every integer  $n \geq 0$ , we have*

$$\sum_{k=0}^n k = \frac{n(n+1)}{2}.$$

*This identity is well known.*

*Proof.* The proof proceeds by induction on  $n$ .

Base case:  $n = 0$  yields  $0 = 0$ .

Inductive step: assume the identity for  $n$ ; then

$$\begin{aligned} (1) \quad \sum_{k=0}^{n+1} k &= \sum_{k=0}^n k + (n+1) \\ (2) \quad &= \frac{n(n+1)}{2} + (n+1) \\ (3) \quad &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

Compare with thm:main and ??; see also ??.

□

**Remark.** *No label here; uses*

$$x^2 + y^2 = z^2$$

*as display.*



FIGURE 1. A simple diagram.